Revealing hidden jump strategies beyond conventional force-plate metrics

Short title: Hidden strategies in jump curves

Article type: Full-length research article (BAICONF 2026 submission BAI2575; IIMB Management Review)

Authors (in this order)

1. Praveen Chougale*
Koita Centre for Digital Health, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India
Postal address: Hostel 12, C-513, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India
Phone: +91 8861755510; Fax: not available
Email: praveenmaths89@gmail.com; ORCID: 0000-0002-5262-4726

2. Usha Ananthakumar
Shailesh J. Mehta School of Management, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India
Postal address: Office 106, Shailesh J. Mehta School of Management, Indian Institute of Technology Bombay, Powai, Mumbai 400076, India
Phone: +91 22 2576 7786; Fax: not available
Email: usha@som.iitb.ac.in; ORCID: 0000-0003-1983-2168

*__Corresponding author:__ Praveen Chougale (details above).

Data source and acknowledgements. The data were collected at PhysioQinesis / Performance Lab, a sports physiotherapy and performance centre in India, between November 2022 and April 2026, and were provided to the authors under a signed non-disclosure and data-sharing agreement between the centre and the authors. The authors thank PhysioQinesis / Performance Lab, its staff and its athletes for access to the data.

CRediT authorship contribution statement. Praveen Chougale: Conceptualisation, Data curation, Formal analysis, Methodology, Software, Visualisation, Writing – original draft. Usha Ananthakumar: Supervision, Writing – review and editing.

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors.

Data availability. The raw force-plate exports are owned by PhysioQinesis / Performance Lab and are covered by the non-disclosure and data-sharing agreement; they cannot be shared publicly. The analysis code is held in the authors' repository (github.com/praveenmaths89/hidden-jump-strategies-forceplate, currently private) and is available from the corresponding author on request; de-identified trial-level data may be made available on reasonable request, subject to the permission of PhysioQinesis / Performance Lab.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work, the authors used AI tools to assist with language editing and code review. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Prior presentation. This paper is the full-paper version of abstract BAI2575, submitted to BAICONF 2026 (Indian Institute of Management Bangalore). The present manuscript reports a reanalysis with documented quality control, and its numbers supersede those of the abstract.

Originality. The authors confirm that (1) the article is their original work; (2) the article, or a significant portion of it, has not been published earlier; (3) it is not under consideration for publication elsewhere; and (4) all copyright requirements for material used directly or indirectly in the article have been met.

Permission for non-public organisational data. Use of the athlete-monitoring data for this research is covered by the signed non-disclosure and data-sharing agreement between PhysioQinesis / Performance Lab and the authors.